

ASSIGNMENT 17

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Subject: Advanced Programming

Import re

Import pytest

Custom Exceptions

Class InvalidEmailError(ValueError):

Def __init__(self, email):

Super().__init__(f"Invalid email address: {email}")

Class UnderageError(Exception):

Def __init__(self, age):

Super().__init__(f"User age {age} is below minimum age requirement of 18")

Registration Service

Class RegistrationService:

```
Def register_user(self, email: str, age: int) -> bool:
```

```
    # Internal assertion
```

```
    Assert email is not None, "Email cannot be None"
```

```
    # Email validation
```

```
    If email.strip() == "":
```

```
        Raise InvalidEmailError(email)
```

```
    Pattern = r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$'
```

```
    If not re.match(pattern, email):
```

```
        Raise InvalidEmailError(email)
```

```
    # Age validation
```

```
    If age < 18:
```

```
        Raise UnderageError(age)
```

```
    Return True
```

```
# ----- PYTEST TEST CASES -----
```

```
@pytest.fixture
```

```
Def service():
```

```
    Return RegistrationService()
```

```
Def test_successful_registration(service):
```

```
    Assert service.register_user(test@example.com, 20) is True
```

```
Def test_invalid_email(service):
```

```
    With pytest.raises(InvalidEmailError):
```

```
        Service.register_user("wrong-email", 20)
```

```
Def test_empty_email(service):
```

```
    With pytest.raises(InvalidEmailError):
```

```
        Service.register_user("", 20)
```

```
Def test_underage_user(service):
```

```
    With pytest.raises(UnderageError):
```

```
        Service.register_user(test@example.com, 15)
```

```
Def test_none_email_assertion(service):
```

```
    With pytest.raises(AssertionError):
```

```
        Service.register_user(None, 20)
```

```
# Optional manual execution
```

```
If __name__ == "__main__":
```

```
    Service = RegistrationService()
```

```
    Try:
```

```
        Result = service.register_user(test@example.com, 18)
```

```
        Print("Registration Successful:", result)
```

```
    Except Exception as e:
```

```
        Print€
```